

Assignment 1

AI-based Village Pond Planning System

Background

Water conservation is a major challenge in rural areas. One effective solution is the construction of ponds at suitable locations to harvest rainwater. Selecting these locations requires analysis of terrain elevation, catchment areas, government land availability, rainfall patterns, and estimated storage capacity.

In this assignment, you will develop a web application that assists village administrators in identifying suitable locations for pond construction using geospatial and satellite data.

Objective

Develop a complete web application capable of analyzing terrain and rainfall information to recommend suitable locations for pond construction. The system should estimate catchment areas, rainfall accumulation, pond depth, and storage capacity, and present the results through an interactive map interface.

Functional Requirements

The application shall support the following features.

1. Display satellite imagery for a selected village.
2. Visualize contour maps.
3. Identify available land suitable for pond excavation.
4. Estimate the catchment area contributing runoff to the selected location.
5. Query historical rainfall data using publicly available APIs.
6. Estimate runoff volume using rainfall and catchment information.
7. Recommend an appropriate pond depth and approximate storage capacity.
8. Overlay all results containing:
 - Selected pond location
 - Catchment area
 - Annual rainfall statistics
 - Estimated runoff volume
 - Recommended pond dimensions
 - Maps and visualizations

Suggested Technology Stack

Students are free to choose their implementation. A suggested stack is

- Python

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- Flask or FastAPI
- OpenCV
- MongoDB/PostgreSQL
- Elevation APIs (OpenZenith, etc.) and Rainfall APIs (IMD, Open-Meteo, NASA POWER, etc.) • A basic frontend library

Deliverables

- Complete source code
- Installation guide
- API documentation
- An accessible Front-end for users
- Final technical report

Important Dates

- **High-Level Design (HLD) Submission:** August 10
- **Prototype Demonstration and Feedback:** Lab Hours
- **Final Submission and Demonstration:** September 5

Evaluation

	Criterion Marks
System functionality	35
Terrain and catchment analysis	20
Frontend and visualization	5
Software design and code quality	15
System Design and Management	15
Documentation and report	10
Total	100